Antonio Balanzategui

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EXPERIENCE

Building Automation System (BAS) Programmer — Applied Statistics & Predictive Maintenance Jan 2026 – Present
Engineered Services, Inc. (contracted to VCU Health) *Richmond, VA*

- Initial February statistical analysis correctly predicted a mechanical change-point from BAS sensor data alone (CUSUM, Mann-Kendall); validated three months later when the EF1 exhaust fan failed catastrophically on March 19, 2026. VCU Health Pharmacy regulatory compliance leadership formally requested quarterly investigations going forward.
- Built and maintain an automated Python pipeline ingesting 10+ months of BAS trend-log data across 12 sensor channels — CUSUM, Mann-Kendall, MEWMA Hotelling T-squared, PCA, Hawkes self-exciting processes, Kalman (RTS) smoothing, Weibull survival, PELT change-point detection — generating the full quarterly investigation end-to-end and replacing manual analysis.
- Decoded Schneider EBO/Continuum RTU control programs from XML export, identifying a disabled proportional humidity reset and a hardcoded exhaust-fan loop bypassing the configured PID — both root causes of ongoing USP 797/800 compliance drift not visible from sensor data alone.
- Diagnosed a BAS historian stale-data cascade (Hawkes branching ratio $n^* = 0.973$, near-critical) affecting all 12 sensor channels; traced it to an ExTL buffer / Maximum Transfer Interval mismatch and implemented a configuration fix that reduced worst-case data staleness from 24 hours to 1 hour (96% reduction).
- Built a MaintainX-to-Freshdesk ticket migration tool using both vendors' REST APIs, automating a re-keying workflow that previously consumed an estimated week of manual work across a 7-person team for hundreds of tickets.

Undergraduate Teaching Assistant, Introduction to GUI Programming Aug 2024 – May 2025
Virginia Tech, Department of Computer Science *Blacksburg, VA*

- Supported 100+ students through object-oriented design, event-driven programming, GUI architecture (layout managers, event listeners, design patterns), and the mathematics underpinning 2D graphics; held weekly office hours and graded projects.

PROJECTS

EvolutionSim — Real-Time Multi-Cell Thunderstorm Simulator 2025 – Present
Personal research project (private repo; demo and code walkthrough available on request)

- Real-time multi-cell thunderstorm simulator coupling 6-species Thompson bulk microphysics, Takahashi non-inductive electrification, and a gauge-invariant Dielectric Breakdown Model (Herrera et al. 2025) on a unified 2.4M-cell GPU fluid grid (Rust + wgpu + WGSL, ~4,600 LoC of compute shaders). Part of a broader ~330K-line platform (Python + Rust) spanning 14 foundation physics modules and 333+ simulation modes.
- Incompressible Navier-Stokes core: Boussinesq buoyancy, Chorin projection, MacCormack semi-Lagrangian advection, Jacobi pressure solve, vorticity confinement; implements schemes from Stam (1999), Selle et al. (2008), and Fedkiw-Stam-Jensen (2001).
- 13K-line Python physics layer (12 modules) drives the GPU; verified numerically against published constants — Heidler return-stroke peak current 29,956 A vs 30,000 A target, breakdown field within 1% per Marshall et al. 1995, gauge invariance and charge conservation verified to floating-point. 124K-line pytest suite covers the broader platform.

De Novo Protein Folding Simulation Spring 2025
CS 4414 (Issues in Scientific Computing), Virginia Tech; team of 4

- Predicted the native structure of a 21-residue peptide (Protein T: NLYIQWLKDGGPSSGRPPPS) from sequence alone — without ML or structural databases — using four custom simulated-annealing protocols in AMBER (Langevin dynamics, generalized-Born implicit solvent); cyclic MSA-MD produced the lowest-energy conformation at -512.26 kcal/mol. **Awarded 2nd Place, VTURCS Undergraduate Research Symposium.**
- Benchmarked five optimization methods (Differential Evolution, Simulated Annealing, Nelder-Mead, Newton's, Steepest Descent) on custom 1D/2D folding-funnel test functions; identified stochastic methods as substantially more reliable on rugged energy landscapes.

EDUCATION

Bachelor of Science, Computer Science May 2025
Virginia Tech, Blacksburg, VA

- GPA: 3.3 | Marvin L. and Leila W. Pollard Scholarship.

SKILLS

Languages: Python, Rust, C++, Java, JavaScript, TypeScript, SQL, C, R, MATLAB; Spanish (proficient).

Statistics & Time Series: CUSUM, PELT, Mann-Kendall, MEWMA Hotelling T-squared, Hawkes processes, Kalman/RTS smoothing, Weibull survival, PCA, anomaly detection.

Scientific Computing & Physics: Navier-Stokes (Boussinesq, Chorin, MacCormack, Jacobi, vorticity confinement), molecular dynamics (AMBER, Langevin, generalized-Born), microphysics, plasma electrification, numerical optimization (simulated annealing, differential evolution, Nelder-Mead, Newton).

GPU & Graphics: wgpu, WGSL, compute shaders, GPU-accelerated simulation, real-time rendering.

ML & Data: NumPy, SciPy, pandas, scikit-learn, matplotlib, Jupyter, data pipelines, feature engineering.

Software & Systems: pytest, REST APIs, Docker, Linux, Git, PostgreSQL, MongoDB, Node.js, Express, React, LaTeX.

BAS & Industrial Control: EcoStruxure, Schneider AS-P / AS-B, Andover Continuum, BACnet/IP, BACnet MSTP, predictive maintenance.